

A Minor Project Report

On

**ENHANCING COMIC SPOILER DETECTION: A
NOVEL DATASET AND ANALYTICAL APPROACH**
*Submitted in the Partial Fulfillment of the Requirements
For the Award of the Degree of*

Bachelor of Technology

In

CSE(Artificial Intelligence and Machine Learning)

By

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DEPARTMENT OF CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

B V RAJU INSTITUTE OF TECHNOLOGY

(UGC Autonomous, Accredited by NBA & NAAC)

Vishnupur, Narsapur, Medak, Telangana State, India – 502 313

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DEPARTMENT OF CSE(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

CERTIFICATE

This is to Certify that the Minor Project Entitled “**Enhancing Comic Spoiler Detection: A Novel Dataset and Analytical Approach**” Being Submitted By

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In Partial Fulfillment of the Requirements for the Award of Degree of Bachelor of Technology in CSE(Artificial Intelligence and Machine Learning) to B V Raju Institute of Technology is Record of Bonafide Work Carried Out During the Period From December 2024 to May 2025 by Him Under the Supervision of

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This is to Certify that the Above Statement Made by the Student are Correct to the Best of Our Knowledge.

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CANDIDATE’S DECLARATION

I Hereby Certify that the Work Which is Being Presented in the Minor Project Entitled “Enhancing Comic Spoiler Detection: A Novel Dataset and Analytical Approach” in Partial Fulfillment of the Requirements For the Award of Degree of Bachelor of Technology and Submitted in the Department of CSE(Artificial Intelligence and Machine Learning), B V Raju Institute of Technology, Narsapur, is an Authentic Record of My Own Work Carried Out During the Period From December 2024 to May 2025, Under the Supervision of Dr. G. Venu Gopal , Assistant Professor and Mrs. Boreda Divya, Assistant Professor.

The Work Presented in this Minor Project Report Has Not Been Submitted By Me For the Award of Any Other Degree of This or Any Other Institute/University.

Velamarthi Karthik [23211A66J0]

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I stand at the culmination of a significant journey, one that has been both challenging and rewarding. The success of my minor project is not solely a reflection of my efforts but a testament to the invaluable support and guidance I have received from many quarters. It is with deep gratitude that I acknowledge those who have made this achievement possible.

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ABSTRACT

Spoiler detection in comics is an emerging field that combines textual and visual analysis to identify content that reveals critical plot points. However, existing approaches are limited in scope, focusing primarily on a narrow range of genres such as romance, battle, and suspense, which restricts their applicability to diverse storytelling styles. This project addresses the challenge of genre limitations by expanding the dataset to include a broader spectrum of genres, such as fantasy, sci-fi, and horror.

A genre-aware classification framework is developed, which integrates textual and visual features to accurately identify spoilers while accounting for genre-specific storytelling elements. Genre-specific annotation guidelines are introduced to improve labeling consistency, and the framework incorporates genre prediction as a feature to enhance spoiler classification accuracy.

By diversifying the dataset and incorporating genre-specific insights, this research enhances the generalizability and robustness of spoiler detection systems, paving the way for broader applications in comic analysis and content moderation.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	ARTIFICIAL INTELLIGENCE
BART	BIDIRECTIONAL AND AUTO-REGRESSIVE TRANSFORMERS
CNN	CONVOLUTIONAL NEURAL NETWORK
DBSCAN	DENSITY-BASED SPATIAL CLUSTERING OF APPLICATIONS WITH NOISE
FER	FACIAL EXPRESSION RECOGNITION
GPT	GENERATIVE PRE-TRAINED TRANSFORMER
GCN	GRAPH CONVOLUTIONAL NETWORK
NLP	NATURAL LANGUAGE PROCESSING
OCR	OPTICAL CHARACTER RECOGNITION
OSNET	OMNI-SCALE NETWORK
RE-ID	RE-IDENTIFICATION
RGB	RED GREEN BLUE
SMOTE	SYNTHETIC MINORITY OVER-SAMPLING TECHNIQUE
TF-IDF	TERM FREQUENCY-INVERSE DOCUMENT FREQUENCY
VLM	VISION-LANGUAGE MODEL
YOLO	YOU ONLY LOOK ONCE (REAL-TIME OBJECT DETECTION ALGORITHM)
YOLOV8	YOU ONLY LOOK ONCE VERSION 8 (LATEST ITERATION OF THE YOLO OBJECT DETECTOR)
XGBOOST	EXTREME GRADIENT BOOSTING

CHAPTER 1

INTRODUCTION

1.1. ENHANCING COMIC SPOILER DETECTION

Comic Spoiler Detection is a new problem in the areas of natural language processing, artificial intelligence, and machine learning. Technological improvements in textual comprehension, sentiment analysis, and content categorization have enabled novel solutions to treat the increased demand for spoiler-aware systems. The fundamental concept of a Comic Spoiler Detection project is to allow machines to recognize and mark possible spoilers in Comic content so that readers can appreciate stories without accidental plot spoilers. It means processing text-usually reviews, social media update, or comment section-to establish if the text has information that discloses key events or conclusions.

Comic Spoiler Detection is a difficult task since it is analyzed on both textual and visual information, as comics integrate narrative text with pictures. It is thereby more difficult in comics than in any other text-based media. The system proposed herein exploits a hybrid approach and employs Natural Language Processing (NLP) to decipher more precise spoiler identification by combining the analysis of both modalities together.

While most current spoiler detection systems are only applicable to popular genres like action or romance, this project expands its boundaries to cover more genres like fantasy, science fiction, and horror. By making use of genre-specific classifiers and enriching the dataset in response, the system is capable of enhancing generalization and maximizing its capacity for accurate detection of spoilers with different emotional forms of storytelling.

1.2. BACKGROUND OF THE DOMAIN

Comics, as opposed to prose, communicate narrative development through a blend of image, spatial panel compositions, and brief textual items such as dialogue or sound effects. The hybrid form comes with challenges unique to automatic content comprehension, particularly for spoiler detection where essential plot information may be communicated visually without direct textual mention. Knowing the semantics of visual narrative — like dramatic reveal, death of a character, or unexpected appearance — needs models that are aware of visual composition and narrative progression, something classic NLP-based spoiler detection can't do on its own.

With the evolution of artificial intelligence, the capability to understand sophisticated, multimodal inputs has also developed. Within the project, OCR technology (Tesseract) retrieves text from panel images to allow language models such as BART to process previously inaccessible dialogue content. At the same time, computer vision models such as YOLOv8 detect and localize human characters, which are then grouped together using DBSCAN to estimate uniqueness and contextual relevance within the panel. This union of vision and language models is an advance over monomodal systems, enabling more complete comprehension and improved generalization across genres and styles of art.

The application of such systems is not merely intellectual but also increasingly business-related. Online platforms for reading comics, fan sites, and content filtering systems might all profit from machine learning-based spoiler removal. As AI deepens its penetration into content delivery streams, the preservation of user experience through prevention of unwanted spoilers can greatly enhance readers' engagement. Therefore, the intersection of vision, language, and machine learning researched here is both timely and meaningful and paves the way for future applications in media moderation, personalization of content, and publishing analytics.

1.3. MOTIVATION

The reason for developing a system such as Comic Spoiler Detection is to achieve a human-like understanding of plot-revealing content across the text and images. Humans can intuitively detect the spoilers in comics by interpreting language, character actions, and visual cues, even across different genres and art styles. This project aims to replicate that capability in machines-enabling them to accurately identify and flag spoilers within comic panels. Doing so will unlock a wide range of benefits, including improved content moderation, enhanced reader experience, and the ability to automate spoiler filtering in digital comic platforms, forums, and social media discussions.

1.4. OBJECTIVES

Accurate Spoiler Detection: This project aims to build a system capable of identifying comic spoilers with a high accuracy diverse comic genre.

Multimodal Content Understanding: The project integrates natural language and computer vision to enable seamless interpretation of textual and visual elements within comic panels, allowing the system to understand narrative context and visual cues.

Application Versatility: The Comic Spoiler Detection system, built using a novel dataset, has potential applications in online comic platforms, forums, moderation tools, and digital archives-helping platforms deliver a spoiler-free reading experience.

1.5. PROBLEM STATEMENT

The problem statement for Comic Spoiler Detection involves the development of an AI model that can automatically detect and classify the spoilers in comic panels. The core challenge lies in bridging the gap between multimodal comic content-visual art, textual dialogue, and contextual storyline elements-and machine-interpretable labels that indicate whether a panel contains a spoiler or not.

The proposed system aims to build a multimodal machine learning model capable of analyzing comic panels using a combination of visual elements, understanding dialogue or narration text through OCR and NLP, and considering metadata such as genre or character recurrence to improve prediction accuracy.

A reliable system should be able to generalize well across various comic genres, character designs, and writing styles. It should also be robust enough to handle image noise, text variability, and complex visual storytelling. By accurately identifying spoiler content, the system will enable the development of spoiler-free comic recommendation engines, intelligent comic reading interfaces, and automated content moderation solutions-thus enhancing the overall reading experience for users.

CHAPTER 2

REVIEW OF LITERATURE

2.1. INTRODUCTION

The study of comic and graphic novel analysis has made tremendous progress over the last few years, with the confluence of computer vision, natural language processing, and multimodal learning being the main drivers. Studies conducted between 2020 and 2025 have emphasized tackling the inherent complexities of comics, including stylized art, intricate panel arrangements, and the fusion of visual and textual data. Deep learning architectures-based techniques such as convolutional neural networks (CNNs), transformer models, and cutting-edge object detectors like the YOLO suite have been extensively used to address tasks from character recognition and re-identification to text extraction and spoiler tagging.

This review covers major contributions in these areas, including means of enhancing OCR performance on comics, multimodal understanding of stories, and building datasets for comic-specific tasks. Additionally, application of vision-language models and transformer-based strategies to genre classification, panel sequence comprehension, and context creation is presented. Lastly, recent breakthroughs in real-time object detection frameworks like YOLOv4 up to YOLOv8 illustrate how rapidly techniques are advancing to support efficient and accurate comic analysis.

Through the investigation of these varied yet complementary studies, this review sets the foundation for comprehending the state-of-the-art of comic analysis tools and establishes avenues for the development of spoiler detection through multimodal fusion of visual and textual information.

2.2. RELATED WORK

[1] Multimodal Analysis for Manga Story Understanding – Matsumoto et al. (2020)

The paper introduces a multimodal framework for understanding manga stories through the combination of facial expression recognition, dialogue analysis, and panel layout parsing. The authors clarify that manga narrative comprehension entails analysis of both visual and textural cues. Facial features are employed to reason about emotional states of characters, while narrative context is contained in dialogue bubbles. The spatial arrangement of panels assists in the comprehension of story flow and transition. The model is tested on a dataset of manga and proves to be better at understanding story segmentation and comprehension. The study of this paper points out the intricacies of manga stories and introduces a joint reasoning method across modalities.

[2] A Deep Multimodal Network for Panel Context Understanding in Comics – Zhang et al. (2021)

The authors set up a deep multimodal architecture that learns the context relationships within comic panels with CNNs for visual features and RNNs for text. They present a dataset of annotated comic panel sequences for coherence and context. The network represents panel-to-panel transitions through the joint consideration of visual scenes as well as embedded dialogue content. This enhances the ability of the system to infer missing panels or predict logical continuations. The work also explores attention mechanisms to find out the most influential features in narrative prediction. The model is tested with retrieval and classification tasks on comic strips.

[3] Improving OCR Performance in Comics Using Domain-Specific Fine-Tuning of Tesseract – Afzal et al. (2020)

The work here addresses the issue of Optical Character Recognition (OCR) in comic text, particularly as a result of stylized fonts and

intricate layouts. The authors optimize Tesseract OCR with a dataset that is chosen specifically from comic speech bubbles and comic narratives. Domain adaptation methods and post-processing fixes are used to enhance recognition accuracy. The model is evaluated on both Western and Japanese comics datasets and achieves considerable performance improvements over typical OCR models. Layout-aware segmentation is also employed to segment text regions more accurately. The study proves that training of a specific domain can significantly enhance OCR performance in visually stylized text documents.

[4] Attention-Guided Scene Text Recognition in Comic Panels – Yuan et al. (2022)

Yuan et al. introduce an attention-guided encoder-decoder model specifically designed to recognize scene text in comic panels. The algorithm is capable of processing irregularly laid-out text, e.g., curved and hand-written speech bubbles. One of the major contributions is the incorporation of spatial attention mechanisms that dynamically focus on text areas at decoding time. The network is trained on a dataset with diverse comic fonts and difficult visual noise. Experimental results indicate enhanced robustness and accuracy over baseline OCR techniques. The solution also features a feature refinement module for improved character-level recognition under cluttered conditions.

[5] YOLOv4-Based Comic Character Detection for Dense Panel Layouts – Wu et al. (2021)

This paper transfers the YOLOv4 object detection model to identify comic characters from dense and visually crowded panels. The authors fine-tune the YOLOv4 model using a comic-specific dataset with human and non-human characters annotated. The system has high-speed detection appropriate for real-time applications. The research also presents bounding box refinement for supporting swollen poses and overlapping objects. Testing on several comic series demonstrates the model's ability to locate characters in affine artistic variations. The

method is traded off between accuracy and computational cost for large-scale panel analysis.

[6] Stylized Character Re-identification in Comics using Deep Feature Embeddings – Jain & Sharma (2023)

Jain and Sharma propose a person re-identification system for stylized comic art employing deep metric learning. The model is trained to generate embeddings that are pose-invariant, outfit-invariant, and style-invariant. OSNet is employed as the feature extraction backbone, suitable for the task of person re-identification. The approach entails pairwise training for reducing intra-character distance and increasing inter-character distance. A labeled character identity dataset in panels is proposed for testing. The results show good performance in following characters through non-linear and visually rich narratives.

[7] BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation – Lewis et al. (2020)

This seminal paper presents BART, a transformer model with both bidirectional and auto-regressive training objectives combined. The architecture is general-purpose for a wide range of generation tasks including summarization, translation, and question answering. BART is pre-trained with a denoising objective that masks text and teaches the model to restore the original. BART does better than current models like BERT and GPT on some benchmarks, particularly on abstractive summarization. There are extensive experiments in the paper on CNN/DailyMail and XSum datasets. Its versatility in both encoding and decoding makes it applicable for downstream tasks in text generation.

[8] Comic Genre Classification using Multimodal Transformers – Gupta et al. (2021)

This work investigates the comic genre classification task based on multimodal transformer models. It integrates visual embeddings from panel images with text embeddings from corresponding dialogues. The

authors employ pre-trained BERT and ResNet models to extract modality-specific features, which are then combined through a transformer encoder. The dataset employed encompasses various genres of romance, action, horror, and comedy. Experiments demonstrate that multimodal inputs significantly surpass text-only or image-only baselines. The work identifies genre-related visual and linguistic patterns that are useful for classification.

[9] Multimodal Spoiler Detection from Text and Metadata in Social Media – Chen et al. (2021)

Chen et al. introduce a multimodal approach to detecting social media post spoilers based on textual information and metadata. The model takes into account linguistic features including sentiment, tense, and named entities, as well as contextual metadata such as timestamps and hashtags. The model is trained using a hand-labeled Twitter dataset for TV shows and movies. Temporal as well as social context are highlighted as key factors in the identification of spoiler content. A vision-language transformer-based hybrid LSTM and attentional network is utilized for the fusion of features. Performance is demonstrated to be better than text-only classifiers.

[10] Webtoon Spoiler Detection via Vision-Language Transformers – Lee et al. (2024)

This work proposes a vision-language transformer model for identifying spoilers in webtoons, a type of digital comics. The model encodes both visual panel information and dialogue text simultaneously with a cross-modal attention mechanism. The authors compile a dataset of webtoon panels tagged with spoiler tags and genre metadata. Pre-trained models such as ViT and BERT are pre-trained and fine-tuned for the purpose. The work shows that collaborative reasoning across images and text highly improves performance in spoiler classification. Attention visualizations are given to read the model's decision-making process.

[11] Multimodal Storytelling Literacy with Comics and Graphic Novels – Adam I. Attwood (2024)

Attwood explores the cognitive and pedagogical dimensions of multimodal storytelling in comics and graphic novels. The research looks at how readers make meaning and synthesize visual and textual components to construct narrative knowledge. Educational implications are presented, illustrating how comics foster literacy and interest in students. The paper examines narrative mechanics like panel sequencing, visual metaphors, and textual anchoring. It borrows case studies from Western comics and manga to illustrate storytelling diversity. The findings stress narrative complexity and interpretive depth necessary for multimodal literacy.

[12] A Deep Learning Pipeline for the Synthesis of Graphic Novels – (2021)

This work presents a deep learning pipeline for generating graphic novel content automatically by incorporating language and image models. Story text is generated using GPT-2, and StyleGAN2 synthesizes visual scenes from the story. The authors come up with a controller that maps textual events into matching visual frames. The controller is guided by semantic tags and character consistency constraints. The model is trained on paired data of story paragraphs and comic strips. The paper shows how AI can be used to help automate aspects of the comic generation process.

[13] One Missing Piece in Vision and Language: A Survey on Comics Understanding – Vivoli et al. (2023)

Vivoli et al. present a thorough review of research studies in the area of comic understanding with an emphasis on the confluence of computer vision and NLP. The paper classifies current methods under tasks such as OCR, panel segmentation, character detection, and story understanding. It surveys more than 100 papers and specifies challenges such as stylized text, non-standard layouts, and narrative

discontinuity. Benchmark datasets and evaluation metrics are also mentioned. Research gaps and future work directions such as multimodal representation learning and temporal modeling are suggested in the survey.

[14] Multimodal Transformer for Comics Text-Cloze – (2023)

This work takes on the task of text-cloze prediction in comics, where the model needs to predict missing dialogue or narration from a panel. A multimodal transformer is introduced, which embeds the visual panel content and the surrounding text. The training is done with comic panels and some of the text boxes masked out. The model is trained to predict the masked content based on contextual information from both modalities. Evaluation is done in terms of accuracy and BLEU scores. Joint modeling is demonstrated by authors as it enhances prediction quality over unimodal baselines.

[15] ComicsPAP: Understanding Comic Strips by Picking the Correct Panel – (2024)

This paper introduces ComicsPAP, a data set and benchmark for assessing panel sequence comprehension in comics. The primary task is choosing the appropriate next panel from a partial strip. Models learn to reason story advancement, visual coherence, and character consistency. The dataset contains a variety of genres as well as Western and Eastern comic book styles. Authors use baseline models with CNNs, transformers, and multimodal fusion networks. Experiment results show that even robust visual-language models fare poorly on inferring subtle narrative hints, leaving space for future work.

[16] Comic Datasets Mix for Detection Benchmarking – Narayan & Kumar (2023)

This work examines the variation in comic object detection model performances when trained and evaluated on different datasets. The authors merge some of the current comic panel datasets and compare

widely used object detection models such as YOLOv4, YOLOv5, and Faster R-CNN. They aim to determine how various models generalize when they are tested using unseen artistic styles and layout structures. By using extensive benchmarking, they expose drastic performance declines when changing datasets based on domain-specific stylizations. The research also talks about the effect of annotation inconsistency and suggests best practices to make more solid benchmarks. By emphasizing dataset diversity as an important consideration, this paper offers insights for future evaluation protocols for comic analysis.

[17] Image Segmentation, Classification, and Recognition Methods for Comics – Ahmed & Liu (2023)

This review paper provides an exhaustive survey of contemporary image processing methods for comics. It discusses segmentation approaches towards panel detection, classification approaches towards comic genres, and object and character recognition models. The authors discuss how the problems of artistic variability, challenging panel layouts, and the presence of mixed modalities pose challenges. They discuss both conventional image processing methods and the latest developments in deep learning techniques such as CNNs and attention-based models. The paper also describes the evolution of datasets utilized for comic image studies. In general, the research provides a necessary guide for researchers venturing into comic-based computer vision tasks.

[18] ComiCap: A VLMs Pipeline for Dense Captioning of Comic Panels – Vivoli et al. (2024)

ComiCap presents a dense captioning pipeline tailored for comic panels with the help of Vision-Language Models (VLMs). The system is designed to produce fine-grained descriptions of a panel's visual content, including character motion, facial expressions, and background activities. The authors modify a transformer-based model

that aligns image regions with synthesized text segments so as to facilitate fine-grained dissection of scene information. They also design a panel-level annotation scheme for training and testing. Results on comic datasets demonstrate that ComiCap can outperform general captioning models. The work fills the gap between general VLMs and domain adaptation captioning tasks in visual narrative.

[19] Manga-MMTL: Multimodal Multitask Transfer Learning for Manga Character Analysis – Nguyen et al. (2021)

This research proposes Manga-MMTL, a transfer learning model designed to address various character-related tasks in manga, such as detection, identity recognition, and emotion classification. Through multitask learning, the model acquires shared underlying representations for all tasks, enhancing generalization on smaller manga-specific datasets. The model combines CNN-based visual encoders and uses task-specific output heads. The authors also refer to the benefits of auxiliary tasks such as facial landmark detection in assisting central goals. Their results on a home-labeled manga dataset show superior performance compared to single-task baselines. The research illustrates the potential for multitask methods in making sense of advanced visual narratives in comics.

[20] YOLOv4: Optimal Speed and Accuracy of Object Detection – Bochkovskiy et al. (2020)

YOLOv4 introduces a series of enhancements to the YOLO (You Only Look Once) object detection framework based on speed and accuracy. The researchers introduce improvements like weighted residual connections, cross mini-batch normalization, and self-adversarial training. All these features collectively allow the model to handle high-resolution images with increased accuracy in real time. YOLOv4 also introduces the utilization of a CSPDarknet backbone, which aids in balancing performance and memory efficiency. Benchmarking

performance indicates YOLOv4 is at the state-of-the-art level for COCO and other public benchmarks. Its module-based design enables fine-tuning it for domain-specific applications, such as artistic and non-natural pictures.

[21] YOLOv5: A New State-of-the-Art Object Detector – Jocher et al. (2020)

The design and performance of YOLOv5, a low-weight but high-powered object detection model by the Ultralytics team, is described in this paper. In contrast to its predecessors, YOLOv5 is implemented entirely in PyTorch, which makes it easier to customize and deploy. It adds functionalities such as learning bounding box anchors automatically, mosaic data augmentation, and built-in model pruning. It is tested on various detection tasks and proves to be incredibly fast with modest resource usage. YOLOv5's modular configuration files also facilitate fast experimentation with varying architectures and hyperparameters. It has since been one of the most used detection models in real-time use.

[22] YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors – Wang et al. (2022)

YOLOv7 extends upon the strengths of previous YOLO models by adding a host of training improvements known as "bag-of-freebies." These consist of scheduled re-parameterization, scaling strategies for models, and compound label assignment for improved localization. The design is optimized for stability in training as well as speed in inference and thus can be deployed on edge devices. YOLOv7 is better in terms of precision as well as FPS compared to YOLOv5 and YOLOv6 on benchmark datasets. The work contains comprehensive ablation studies to attribute the contribution of every training enhancement. It creates a new gold standard in trading off real-time performance with detection accuracy.

[23] Real-Time Object Detection with YOLOv8: Architecture and Performance – Ultralytics Team (2023)

YOLOv8 presents an innovative architecture that prioritizes flexibility and modularity for object detection. Notable improvements include a classification and regression decoupled head, enhanced anchor-free mechanisms, and multi-input resolution support. The model is optimized with up-to-date methods such as EMA (Exponential Moving Average) and mixed-precision training to accelerate convergence. The article offers benchmark comparisons that depict YOLOv8's enhanced performance in the categories of small, medium, and large object detection. Its open-source publication comes with Roboflow integration that makes dataset handling and model fine-tuning easier. YOLOv8 presents itself as the most versatile YOLO version to date.

[24] Deep Learning Techniques for Comic Panel Segmentation and Classification – Li et al. (2021)

This work investigates comic panel segmentation, or the detection and separation of individual panels from a page structure. The authors use a CNN model trained on labeled comic pages with diverse panel configurations. Their model has edge detection and contour analysis to handle irregular borders and overlapping frames. Besides segmentation, they also propose a panel classification module to mark each panel according to its narrative function (e.g., action, dialogue, transition). Testing on mixed-style comics presents high accuracy in both segmentation and panel classification. The work highlights the importance of structural layout understanding for comic analysis.

[25] Multimodal Fusion for Comic Scene Understanding – Nguyen et al. (2022)

This study suggests a multimodal fusion system to enhance the understanding of scenes within comic panels. The system processes both visual input (panel images) and textual input (speech bubbles or

captions) and fuses them through a late-fusion approach. Deep CNNs encode images, while transformer models encode text. The fusion is performed at the decision level so that the two modalities can be processed separately prior to their combination. The model is tested on tasks such as genre classification and emotional tone identification. Outcomes show that the fusion strategy produces more context-sensitive predictions compared to unimodal systems.

[26] Transformer-based Models for Comic Dialogue Understanding – Wang & Liu (2023)

The paper discusses the potential use of transformer models for comic dialogue parsing and interpretation. Due to the discontinuous and visually embedded nature of comic text, conventional NLP approaches frequently fail. The authors train BERT and its variants on comic dialogue corpora with attention to tasks such as sentiment and intent classification. Additional preprocessing procedures are used to maintain the hierarchy of speech bubbles and speaker turns. The outputs of the model are utilized to reconstruct sequences of dialogue and trace character interactions. Experimental results demonstrate that when properly adapted, transformers can successfully model the subtleties of conversational text in comics.

[27] Recent Advances in OCR for Manga Text Extraction – Chen et al. (2020)

Chen et al. study the particular challenges of text extraction in manga, including non-standard text orientations, curved baselines, and irregular fonts. They suggest a hybrid OCR system that integrates image pre-processing, deep segmentation, and language model correction. The system includes a font-aware convolutional backbone that enhances recognition on stylized text. A post-processing layer based on a character-level language model assists in the correction of incomplete or uncertain predictions. The model is evaluated on a multilingual

manga data set and demonstrates considerable improvement over general OCR software. The paper offers valuable information on applying OCR to visually sophisticated comic material.

[28] Character Detection in Comics Using YOLOv5 and Data Augmentation – Kumar & Singh (2021)

This work modifies the YOLOv5 object detection model to detect characters in comic panels with emphasis on enhancing generalization by means of data augmentation. Data augmentation strategies such as cutmix, color jittering, and geometric distortion are used to mimic differences in artwork style and panel layout. The model is trained from a dataset of diverse comic characters from different genres. Testing results illustrate that augmentation greatly improves performance on novel panels. The authors also compare YOLOv5 with other detection models and have positive speed-accuracy tradeoffs. Their method gives a real-world solution to scalable character recognition.

[29] Clustering Comic Characters Using DBSCAN and Deep Feature Embeddings – Patel & Sharma (2023)

This paper presents a method for the identification of unique comic characters using clustering methods on deep visual embeddings. The authors use a CNN to obtain high-dimensional features from character detections and use DBSCAN to cluster visually similar instances. The clustering assists in separating repeated occurrences of the same character in different panels. The approach does not depend on labeled identity data, so it is appropriate for unsupervised situations. The approach is demonstrated to have high purity clusters when applied to a varied comic dataset. The method is particularly effective for character frequency and interaction analysis across large story arcs.

[30] Comic Genre Classification with Multimodal BERT Models – Gupta et al. (2021)

The authors, in this paper, introduce a multimodal BERT-based genre classification for comics. Visual features are handled with CNNs, and speech bubble textual information is represented using BERT. These two modalities are combined at the embedding level and fed to a classifier trained to separate between genres such as action, romance, horror, and fantasy. The work incorporates experiments that demonstrate the benefits of modality interaction over independent pipelines. The authors further visualize attention maps to understand which features have the most impact on genre prediction. The research proves the feasibility of using BERT-based fusion models for classification tasks.

[31] Visual-Textual Coherence Modeling in Comic Panels – Lee and Park (2022)

Lee and Park are interested in semantic alignment of images and dialogue text in comic panels. The model employs a dual-branch encoder framework, where one branch encodes visual panel and the other encodes embedded dialogue content. They train the model using contrastive learning to align panels with their appropriate texts and differentiate them from inappropriate samples. The work presents a dataset of comics with annotated panel-text mismatches and supports supervised training for coherence scoring. Recall@k and cross-modal retrieval accuracy are employed as metrics for evaluation. Improvement in narrative flow understanding through image-text matching is demonstrated.

[32] Detecting Out-of-Context Panels in Comics Using Vision Transformers – Kwon et al. (2023)

This work suggests an approach to identify panels that are misplaced or break the narrative flow in comics. The authors use Vision Transformers (ViTs) for learning abstract visual semantics and identifying discontinuities where a panel breaks the narrative

continuity. The model is trained on carefully selected comic sequences with deliberately misplaced panels. Contextual information like orientation of the face, consistency of the background, and alignment of the layout is utilized to evaluate continuity. The model achieves high performance on panel ordering and anomaly detection tasks. This work contributes to machinery for ensuring story coherence in automated comic analysis or generation.

[33] Comic Layout Understanding using Graph Neural Networks – Bae et al. (2021)

Bae et al. introduce a graph-based method to comprehend the spatial layout of comic structures. All panels are considered as nodes, and spatial or narrative connections between panels are represented as edges. The model, with a Graph Neural Network (GNN), learns to predict panel reading order, speech bubble directions, and general layout composition. The dataset contains a broad variety of styles, ranging from conventional manga to Western strips. The model is compared against heuristic-based ordering strategies and demonstrates more accuracy in logical reading path prediction. The method is particularly useful in determining non-standard or unusual comic structures.

[34] Dialogue-Driven Comic Panel Retrieval – Tanaka and Saito (2022)

This paper introduces a retrieval model that identifies comic panels from a given dialogue input. A bi-modal embedding framework is created, wherein text is represented with a language model and panel images are represented with a CNN. Embeddings are matched in a common space with contrastive loss, allowing effective retrieval. The data used to train comes in the form of dialogue-panel pairs mined from digital comic archives. Top-k metrics are used by the paper to measure retrieval accuracy and show that adding context windows (adjacent dialogue) greatly improves performance. The approach can be used in web search and web comic recommender systems.

[35] Emotion Recognition in Manga by Facial and Bodily Expressions – Hayashi et al. (2021)

This work deals with emotion recognition from manga using facial expressions, posture, and visual cues based on context. An in-house dataset is annotated with emotional labels based on Ekman's emotion taxonomy. The authors train a hybrid CNN-RNN model to learn spatial and temporal features so that it can represent small emotional changes between panels. Attention layers are employed to highlight the concerned facial or corporeal areas. Experiments demonstrate high accuracy in detecting nuanced emotional states like embarrassment or sarcasm, which are frequently overdrawn in manga illustrations. The research contributes to affective computing in stylized visual media.

[36] Modeling Narrative Arc in Comics with Multimodal Encoders – Shen et al. (2023)

Shen et al. investigate modelling narrative arcs based on sequences of comic panels. The system, proposed by them, employs multimodal encoders to represent visual progression, dialogue flow, and character interactions. They establish narrative arc stages (e.g., setup, conflict, resolution) and annotate a dataset as such. Applying sequence modelling with attention, the model learns to divide a narrative into stages of a story. Both synthetic and natural comic strips are used to test the model. Results show how it can generalize among genres and assist in organizing convoluted narratives into meaningful units.

[37] Speech Bubble Localization in Manga Using Edge-Aware CNNs – Lin et al. (2021)

Lin et al. present a dedicated edge-aware convolutional network to recognize speech bubbles in manga, whose borders tend to be stylized or suppressed. The model combines an edge detector with a semantic segmentation backbone to identify bubble areas from background text

and artwork. The system is trained on labelled manga pages of varied bubble shapes, including jagged and floating styles. Evaluation is done using IoU and pixel accuracy measures, demonstrating better performance compared to baseline segmentation models. The work tackles the challenge of bubble detection when common bounding box assumptions are not met.

[38] Automatic Comic Strip Summarization through Panel Importance Ranking – Zhao and Liu (2020)

The article proposes a system to summarise comic strips by choosing the most relevant panels. Importance is represented by a compound of visual salience, dialogue density, and plot progression hints. A ranking model is learned through supervised learning on annotated summaries from human readers. The authors introduce a summary generation algorithm that strikes a balance between plot coverage and diversity. The system is tested with ROUGE scores and user studies, demonstrating that machine-generated summaries closely match human preference. This approach can assist with content browsing, previewing, and archival classifying.

[39] OCR Challenges in Stylized Text Recognition in Comics – Yamamoto et al. (2022)

Yamamoto et al. examine the special challenges of OCR on stylized comic text, such as decorative font, skewed baseline, and overlapping features. They compare several OCR engines and identify typical failure cases. The authors then suggest a hybrid approach that is a CNN-based detector paired with a sequence-to-sequence recognizer that is fine-tuned on comic-style fonts. They pose preprocessing methods like font standardization and layout normalization to enhance recognition rates. Their experiments demonstrate how slight modifications to OCR pipelines can greatly enhance accuracy for comic-specific information.

[40] Comic Strip Coherence Evaluation with Visual-Linguistic Entailment – Singh and Mehta (2023)

This study proposes a new task of evaluating the coherence of comic strips using a visual-linguistic entailment framework. The idea is to assess whether a sequence of panels logically follows from one another, much like textual entailment in NLP. The authors adapt a multimodal transformer model that inputs panel images and accompanying dialogue, then outputs a binary decision indicating coherence. A new dataset of incoherent and coherent panel sequences is presented. Multimodal entailment models are shown to outperform unimodal baselines through experimental results. This task enables new applications in comic creation and curation quality control.

[41] Comic Scene Transition Classification Using Temporal CNNs – Rao et al. (2021)

Rao and colleagues build a model for scene transition classification in comics, e.g., moment-to-moment, action-to-action, or scene-to-scene transitions. Their method employs temporal convolutional networks that analyze sequences of panels and derive features from imagery as well as embedded text. Transition types are categorized based on Scott McCloud's typology. The model is hand-labelled and assessed with accuracy and confusion matrix analysis. The authors explain how various genres have distinct patterns of transitions. Their work aids in the comprehension of narrative pacing and structure in visual narratives.

[42] End-to-End Comic Character Tracking with Siamese Networks – Zhou et al. (2022)

This work introduces an end-to-end comic character tracking system based on a Siamese network architecture. The network matches candidate character regions between panels to detect repeated occurrences of the same character. Feature embeddings are learned to

be pose-, outfit-, and artistic style-invariant. The system incorporates a re-identification module to associate detections across long sequences. Training is done on an annotated character identity dataset for whole volumes. The model shows robustness for character tracking even with extreme artistic differences. Applications are automatic character tagging and storyline analysis.

[43] Dialogue-Aware Panel Segmentation for Webcomics – Kaur and Desai (2021)

Kaur and Desai present a segmentation approach that leverages dialogue content to assist in identifying panel boundaries in webcomics. In contrast to printed comics, webcomics tend to have non-regular layouts and overlapped components. The authors employ a combination of image segmentation and natural language components to cue the panel detection process. Positions of speech bubbles and reading order hints are included as auxiliary inputs to the model. They test their system on a carefully curated dataset of contemporary webcomics and achieve improvements in boundary detection accuracy. The technique is very effective for comics with speech-led layouts.

[44] Fine-Grained Comic Emotion Classification with Joint Visual-Text Encoding – Choi et al. (2022)

This paper addresses the fine-grained emotion classification in comics where emotions tend to be implied rather than shown directly. The authors create a joint visual-text encoder that combines panel art and related dialogue into a common representation. Emotions are classified into fine-grained categories such as anticipation, disgust, and awe. Contrastive learning is employed by the model to match emotionally similar examples in the embedding space. A new multi-label emotion-tagged dataset is proposed to be used for evaluation. The work registers high F1 scores and emphasizes the richness of emotional subtlety in illustrated media.

[45] Self-Supervised Learning for Comic Panel Layout Understanding – Mukherjee and Banerjee (2023)

Mukherjee and Banerjee present a self-supervised learning method to learn panel layout without using labelled data. The technique employs the pretext task of predicting the correct ordering of randomly shuffled panels to learn effective layout representations. The model utilizes a transformer-based encoder, which accepts panel embeddings as input and outputs an ordering score. Contrastive losses and permutation prediction objectives are used to train the model. The authors demonstrate that the learned embeddings generalize to downstream tasks such as reading order prediction and panel classification. The work proves the feasibility of learning structure-aware features with little annotation.

[46] GAN-based Generation of Comics from Scripts – Nishida et al. (2021)

The authors introduce a generative adversarial network that transforms narrative scripts into comic panels. The model comprises a layout planner and a visual generator, which is trained on paired script-image data. The system manages character placement and rudimentary scene composition.

[47] Manga Face Detection using Multi-Scale Anchor-Free Models – Tokui and Watanabe (2022)

This paper proposes an anchor-free manga face detector that can deal with drastic variations in scale and style. The model utilizes feature pyramid networks to enhance the localizability. The model beats anchor-based models both in terms of speed and accuracy.

[48] Text-Visual Retrieval in Comics using Cross-Attention Networks – Jeon et al. (2022)

Jeon et al. employ cross-attention mechanisms to align dialogue text with the visual elements in comic panels. The model has bidirectional retrieval tasks and achieves improvements in panel-to-text and text-to-panel matching. Training is conducted on a large comic dataset.

[49] Automatic Character Naming in Comics Using Face Clustering and Co-Reference Resolution – Zhang et al. (2021)

This paper introduces a system which names and identifies comic characters through face embeddings clustering and disambiguating textual references. The system integrates visual and linguistic cues in connecting names with faces across panels. The method has high name assignment accuracy.

[50] End-to-End Comic Story Generation with Multimodal Transformers – Rao and Lin (2023)

Rao and Lin introduce a multimodal transformer that generates comic stories from sketches. The model handles plot structure, creates dialogues, and produces corresponding visual scenes with the help of GANs. It shows story consistency and visual-textual correspondence.

[51] Panel Captioning in Webcomics Using Dual Encoder Networks – Sharma et al. (2022)

This paper introduces a double encoder model for comic panel caption generation by encoding text and image features. The model is trained to generate descriptive, coherent sentences and is assessed based on BLEU and METEOR scores.

[52] Masked Visual Modelling for Comics Pretraining – Li et al. (2023)

The authors use masked modelling methods to pretrain visual encoders on comic images. The model learns structure and style information beneficial for downstream tasks. There are apparent advantages in classification and detection performance through pretraining.

[53] Speech Bubble Shape Analysis for Style Identification in Comics – Zhou and Han (2022)

This paper categorizes comics by examining the form of speech bubbles and their style. Geometric attributes are extracted and applied to style clustering. Results show that design derives reliably from bubble design to specify artistic genre or era.

[54] Cross-Language Comic Translation Using Visual Grounding – Wu et al. (2021)

Wu et al. create a visual grounding-based comic translation system for enhancing the quality of translation. The model aligns translated text with visual context to ensure that meaning is maintained. Evaluation is done with both automatic metrics and human evaluation.

[55] Adversarial Examples in Comic OCR – Singh and Joshi (2022)

This paper investigates the vulnerability of OCR systems on comic books to attacks. The paper demonstrates how minor perturbations in font or layout of comics can drastically lower the accuracy of OCR. It is a concern for OCR robustness on creative media.

[56] Modelling Hummy in Comic Dialogues Using Sentiment Flow – Nakamura et al. (2021)

Nakamura et al. study hummy in comics using sentiment changes between dialogue turns. They construct a punchline detector classifier that identifies punchlines by predicting sentiment transitions. It is evaluated on webcomic joke-tagged datasets.

[57] Joint Text-Image Embedding for Comic Panel Alignment – Liu and Chen (2022)

This work suggests a model that places comic images and text in the same latent space. The system gets trained for mapping similar pairs in

terms of triplet loss. It facilitates a number of tasks such as retrieval as well as panel-text matching.

[58] OCR-Free Dialogue Extraction from Comics Using Text Region Segmentation – Ahmed and Kim (2023)

Ahmed and Kim propose a way of dialogue extraction without OCR by parsing and segmenting text areas directly. The system is efficient in the case of stylized fonts and noisy backgrounds, particularly in manga.

[59] Panel Sequencing in Comics Using Graph-Based Constraints – Chatterjee et al. (2021)

This work introduces a graph constraint-based panel ordering system based on layout and narrative hints. It estimates reading flow and spatial interactions to arrive at correct orders. It is more accurate than heuristic-based approaches.

[60] Unsupervised Comic Panel Clustering Using Layout and Style – Tran and Hoang (2022)

This work groups panels into sequence segments according to visual appearance and layout similarity. It employs unsupervised feature extraction and clustering models to detect scene boundaries. The method works effectively even in the absence of labeled data.

[61] Stylized Text Enrichment for Comic OCR – Wei et al. (2023)

Wei et al. address the issue of low OCR accuracy on stylized comic fonts by proposing a pre-processing pipeline that enriches text quality prior to recognition. The approach has font normalization, stroke enrichment, and background noise removal, with the method applying image filtering and GAN restoration to produce cleaner text areas. Their pipeline notably enhances the recognition rate of vanilla OCR engines such as Tesseract and EasyOCR. The model is tested on a varied collection of comic styles, proving to be robust across a range of visual

complexities. The task tackles a significant bottleneck in comic digitization and text extraction.

[62] Automatic Genre Tagging of Comic Panels via Multimodal Features – Batra and Singh (2022)

This work introduces a genre classification system relying on both visual information from panels and textual features from dialogue. The system adopts a late-fusion approach to integrate CNN-based visual representations and BERT-based textual features. Horror, romance, and comedy are forecast by a multilabel classifier trained over a curated dataset. F1 score and precision-recall over each genre label are used as evaluation metrics. Results indicate that the combination of both modalities makes predictions more accurate compared to using either one independently. The work identifies the ability of genre-aware tagging to be used for recommendation and organization.

[63] Image-Only Comic Strip Summarization – Yamada and Suzuki (2021)

Yamada and Suzuki introduce a scheme for image-only comic strip summarization, independent of text. The system rates panels in accordance with visual salience, action intensity, and character prominence. These attributes are utilized to choose a representative subset of panels that transmit the overall narrative. The method is particularly helpful in the case of silent comics or situations where dialogue is absent or unnecessary. Human judgments indicate that the summaries generated preserve important narrative information. The research helps to provide accessibility, particularly in text-light or multilingual comic formats.

[64] Speaker Attribution of Dialogue in Comics using Character Matching – Jain et al. (2022)

Jain et al. present a system for assigning dialogue bubbles to the appropriate characters in comic panels. They employ a combination of facial recognition and spatial distance to determine speakers, dealing with sophisticated layouts and overlapping characters. A matching algorithm assesses every candidate character in terms of distance to the bubble, visual attention, and past dialogue context. They train and test on speaker-labeled annotated comics. Experimental outcomes indicate improvements in the accuracy of attribution, particularly in multi-speaker environments. This work is critical to automating reading order and dialogue parsing for comics.

[65] Context-Aware OCR Correction in Comics – Patel and Deshmukh (2023)

The work in this paper focuses on the post-processing of OCR results from comics and typically contains misread characters and punctuation because of stylized fonts. Patel and Deshmukh employ a contextual language model that examines the structure of dialogue and corrects probable errors. They employ a fine-tuned transformer-based model which predicts corrections based on character-level context. It is trained using a supervised approach with a dataset of OCR output as well as manually corrected text. The method dramatically lowers error rates, especially for slang-ridden or colloquial speech. The research enhances the accuracy of downstream processes such as text analysis and translation.

[66] Tracking Visual Themes Across Comic Volumes – Seo and Hwang (2022)

Seo and Hwang introduce a method of discovering repeated visual motifs across comic issues or volumes. The system uses feature extraction via CNNs and clustering methods to categorize panels according to color palettes, background objects, and symbolic symbols. It can identify visual patterns such as repeated flashbacks, dream

sequences, or character motifs. The technique is tested on curated datasets of serialized comics. The technique is particularly beneficial for content indexing, narrative structure analysis, and fan-tagging. The research investigates how persistent visual storytelling patterns change over time.

[67] Speaker Identification in Multi-Character Comic Panels – Rana and Mohanty (2021)

This paper centers on the identification of speaker in panels with multiple characters and entwined dialogues. Rana and Mohanty propose a model that takes into account visual distance to speech bubbles, character direction, and story coherence. The model incorporates a spatial analysis engine and dialogue history tracker to minimize ambiguity. The model is trained on comic scenes with intricate conversational structures. Results of evaluation demonstrate better speaker detection compared to heuristic baselines. This work is useful for translating visual comics into machine-readable dialogues.

[68] Visual Text Fusion for Comic Emotion Detection – Fang et al. (2022)

Fang et al. create a multimodal emotion detection framework for comics by combining image and text representations. A CNN extracts visual emotion signals from the artwork, while a transformer model encodes the text sentiment. Both modalities are fused at the fusion layer and output a fine-grained emotion class. The authors label a dataset with several emotion labels per panel, such as joy, fear, sadness, and surprise. The model is accurate at detecting slight emotional changes. This piece of work adds to affective computing for visual narrative.

[69] Comic Panel Denoising Using Image Translation Networks – D'Souza et al. (2023)

The paper introduces an image-to-image translation model that is trained to denoise visual artifacts from scanned or vintage comic

panels. The model employs a U-Net-like architecture to denoise, restore color, and sharpen lines. A synthetic-noise training set is established through degrading clean panels to mimic wear and tear. The panel is restored from the system with little distortion to artistic details. PSNR and SSIM measures are used for evaluation, and user studies affirm enhanced readability. The tool is designed for archival restoration and digital remastering of comic books.

[70] Explainable AI for Comic Scene Classification – Khanna and Roy (2022)

Khanna and Roy suggest a comic scene classifier that combines explainability through visual attention maps. The algorithm classifies panels into types of scenes (e.g., action, conversation, transition) and marks the visual and textual features responsible for the prediction. Through Grad-CAM and attention visualization methods, the model provides insights into its reasoning. This transparency is also helpful for debugging and model trust improvement. The classifier is trained on annotated comic scenes and tested on vintage and modern series. It delivers prediction and interpretation in the same framework.

2.2. RESEARCH GAPS

1. Limited Work on Spoiler Detection in Comics.

The majority of the work on spoiler detection is on text-oriented websites such as social media, with comics which are very visual and story-centric relatively unexamined.

2. Insufficient Multimodal Feature Integration

While some work integrates visual and text-based data, few effectively combine features such as character presence, dialogue sentiment, genre, and scene context in unison for tasks such as spoiler classification.

3. Generic Object Detection

Models Performed Poorly on Comic Stylization Standard YOLO models are good on natural images but tend to fail when directly applied to stylized comic art. There are not enough fine-tuned models for comic character and scene detection.

4. Insufficiency of Character Tracking Across Panels

Continuity and identity tracking of characters across panels is still a difficult task. Few papers try re-identification in stylized art forms such as manga or western comics.

5. Text Understanding in Stylized Fonts Remains Fragile

Even with advances in OCR tools, text extraction from speech bubbles, particularly in highly stylized or handwritten fonts, remains prone to errors affecting downstream tasks such as summarization.

6. Lack of a Single Pipeline for Spoiler Detection

No publicly available pipeline exists that can take a comic panel, pull out relevant visual and textual features, and then classify it for spoiler content a task this project seeks to fill.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

In recent years, Artificial Intelligence and deep learning methods have been utilized increasingly to interpret and draw narrative meaning from visual storytelling media including comics and graphic novels. A new challenge in the novel is detecting "spoiler" material — the panels that expose important plot twists, character deaths, or climactic action. Manual labelling of such material is unwieldy at scale due to the intricate interaction of visual, textual, and contextual factors.

This research introduces a multi-modal, feature-based framework for comic panel spoiler classification that combines visual object detection, character-level clustering, natural language processing, and machine learning. The system uses YOLOv8 for character detection, DBSCAN for estimating novel character count, BART for text summarization and captioning, and XGBoost as the final classifier model. The method is interpretable, modular, and domain-adaptable over different comic genres.

3.2 METHODOLOGY FOR SPOILER CLASSIFICATION IN COMIC PANELS

The steps for creating the spoiler classification system are illustrated below. Each aspect is considered as part of an n-stage pipeline including panel image processing, text extraction, semantic comprehension, and final classification. The system is trained and tested on a home-created dataset of 2275 labeled comic panels.

3.2.1 Panel Collection and Labelling

The dataset includes 2275 comic panels that were manually selected from a variety of comic series. Every panel is tagged as either a spoiler

or non-spoiler. A panel is tagged as a spoiler if it involves significant plot revelations like:

1. Death or transformation of a key character
2. Revealing secrets or relationships
3. Unplanned alliances or betrayals
4. End-of-arc showdowns
5. The labelling was done by domain-aware human annotators in order to provide high-quality ground truth.

3.2.2 YOLOv8-based Character Detection

The initial visual feature extracted from every panel is the number and presence of characters. For this purpose, a fine-tuned, in-house YOLOv8 object detection model was utilized. This is trained on comic-specific annotations to identify human characters, even when found in outsized postures and cartoonish art.

- The YOLOv8 model provides bounding boxes for all identified characters in every panel.
- These detections are stored as coordinates and then sent to the next phase for clustering.



Fig. 3.2.2 Yolov8 Detection

3.2.3 DBSCAN-based Unique Character Estimation

As opposed to images in the real world, comic panels sometimes contain repeated occurrence of the same character within a panel (e.g., motion of the same character or reflections). To prevent overcounting, I use

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) to cluster overlapping or adjacent detections.

- The spatial location of bounding boxes is the feature utilized in clustering.
- DBSCAN determines separate clusters of character detections in order to approximate the number of individual characters per panel.
- This count of individual characters is utilized as a numeric feature for spoiler prediction based on the assumption that panels with higher counts of individual characters are likely to be climactic or important.



Fig. 3.2.3 DBSCAN Unique Character Estimation

As shown in the figure 3.2.2 and 3.2.3, yolov8 treats each detection as a unique character, using DBSCAN the same characters will be clustered and provided their respective id's and will be treated as a single character and provides us with the unique character count.

3.2.4 Text Extraction using Tesseract OCR

- Speech bubbles and text of stories in panels hold crucial information. To get this, the Tesseract OCR engine is invoked:
- Raw panel image is preprocessed by OpenCV for contrast boost and background noise reduction.

- Language-specific configuration (mostly English) is used to invoke Tesseract.
- Extracted text is saved and purged of artifacts (e.g., newline breaks, font effects).
- This text serves as the foundation for semantic feature generation in subsequent steps.

3.2.5 Caption Generation using BART Summarization

- Comics usually include redundant or wordy text that can hide the main event. To generate a brief description of what a panel is "about," I apply the BART summarization model (Facebook AI):
- The raw text extracted from Tesseract is fed into the BART transformer.
- BART produces a one-line summary caption that captures the main event or situation.
- Example: "I can't believe he betrayed us!" → "Character betrayal revealed"
- This generated caption assists the model in concentrating on core narrative actions instead of surface-level dialogue.

3.2.6 Genre Classification with BART

- To put a panel's probability of being a spoiler into context, I treat genre as a categorical feature. Action and thriller genres, for example, are more likely to contain spoiler-filled content.
- I re-use the BART model in classification mode to predict the genre of a panel based on its extracted text.
- Genres covered: Action, Romance, Mystery, Fantasy, Slice-of-Life, etc.
- The forecasted genre is used as an extra feature for the XGBoost classifier.

3.2.7 Construction of Feature Vector

- Every panel is ultimately represented by a structured feature vector made up of the following features:

Table 3.2.7 Feature Vector for Spoiler Classification

Feature	Type
Unique Character Count	Numeric
Genre	Categorical
Caption Embedding	Text (Vectorized)
Raw Text Summary	Text (Vectorized)

All text features are vectorized via TF-IDF or sentence embeddings, and categorical features are one-hot encoded. The final feature vector is fed as input to the ultimate classifier.

3.2.8 XGBoost Spoiler Classification

- The last step consists of feeding extracted and encoded features into an XGBoost classifier, selected due to its high accuracy, interpretability, and efficiency for tabular data.
- The classifier is trained with an 80-20 train-test split.
- Hyperparameter tuning is performed with cross-validation and grid search.
- The evaluation metric is Accuracy, Precision, Recall, and F1-Score.
- XGBoost is ideally suited to this task since it can deal with mixed data types (numeric, categorical, and vectorized text) and model non-linear feature interactions well.

3.3 Dataset Overview and Preparation

The quality and preparation of a dataset are the building blocks of any machine learning project. For this comic panel spoiler classification system, a dataset of comic panels was carefully prepared to support efficient training, validation, and testing of the model.

3.3.1 Dataset Description

The dataset comprises 2275 labeled comic panels drawn from varied comic books and graphic novels. Each panel is marked as either spoiler or non-spoiler depending on the potential of its content to disclose important plot points.

Important features of the dataset are:

- Panel Images: High-resolution RGB images capturing different art styles, character setups, and backgrounds.
- Text Content: Interspersed dialogues and narration within panels, which differ in font, size, and position.
- Annotations: Spoiler tags added by manual expert judgements to confirm high-quality ground truth.
- The dataset contains a wide variety of comic genres and art styles, making models robust across a wide range of scenarios.

3.3.2 Dataset Preparation

For the preparation of the raw panel images prior to model consumption and feature extraction, the following steps were carried out for preprocessing:

- Image Normalization: The images were resized and normalized to standard sizes for detection and OCR consistency.
- Text Extraction Preprocessing: Images were first converted to grayscale before Optical Character Recognition (OCR) using Tesseract for enhanced text detection accuracy.
- Text Cleaning: Cleaned text was subjected to noise cleaning such as removal of special characters, fix of OCR misreads, and text casing normalization.
- Character Detection Annotations: Manually annotated character bounding boxes were used to fine-tune YOLOv8 to realize high detection accuracy.

- Clustering for Unique Character Count: After YOLOv8 detection, DBSCAN clustering removed duplicate detections to approximate the unique character count per panel.
- Caption Generation Preparation: Cleaned text was aggregated and fed to a BART summarization model to produce brief scene captions.

3.3.3 Class Imbalance Handling

The data in this research was highly skewed in terms of class distribution, with the spoiler panels being highly underrepresented in relation to the non-spoiler panels. Learning from this biased data can lead to an overly strong bias toward the majority class at the expense of the model's performance in correctly identifying spoilers.

In order to counter this effect, the Synthetic Minority Oversampling Technique (SMOTE) was utilized as part of the data preprocessing pipeline.

Overview of SMOTE :

SMOTE is a sophisticated oversampling method that creates synthetic samples of the minority class (here, spoiler panels) by interpolating between current minority samples. In contrast to random oversampling, which merely duplicates samples and can cause overfitting, SMOTE generates new, varied data points in the feature space.

Application in My Pipeline

Input Features to SMOTE:

The last feature vectors (following character count estimation, OCR text summarization, and genre classification) were fed to SMOTE for class distribution balancing.

Steps Taken:

- Identified the minority class as the spoiler.
- Used SMOTE with the imblearn library in Python.
- Created synthetic data points for the spoiler class in feature space.
- Added to the original dataset to form a balanced training set.

- Effect on Model Performance
- Better Recall of Spoiler Class: The model was made more attuned to the minority class, lowering the number of false negatives.
- Balanced Learning: The classifier was trained using an equal number of spoiler and non-spoiler instances, which enhanced its generalization.
- Increased Robustness: Application of SMOTE helped enhance the edge cases and infrequent spoiler occurrences.

3.4 SYSTEM ARCHITECTURE AND PIPELINE

The architecture for comic spoiler detection is implemented as a multi-stage, modular pipeline that works on visual and textual features of comic panels to classify whether a panel has a spoiler or not.

The system starts with the input image of the comic panel, which is initially run through a fine-tuned YOLOv8 model specifically trained for identifying human characters in comic drawings. The produced bounding boxes of the characters are then clustered with the DBSCAN algorithm to provide an estimation of distinct characters, allowing character-based context and character interactions to be captured in the panel.

Following this, the textual content (for example, captions, dialogues) within the panel is extracted through the use of Tesseract OCR, allowing the narrative content to be analyzed.

The text is then passed through BART, a transformer model, to generate a short summary of the content of the panel as well as classify its genre (e.g., action, romance, drama). These results—character length, summarized content, and genre—are aggregated into a feature vector representing the multi-modal context of the panel.

Prior to training the final classifier, SMOTE (Synthetic Minority Oversampling Technique) is utilized to address class imbalance in the training set by synthesizing examples of the minority (spoiler) class.

Lastly, the feature vector is passed through an XGBoost classifier, which gives a prediction on whether the panel is a spoiler or not.

The modularity of the architecture makes interpretability straightforward, model substitution flexible (e.g., substituting BART with GPT-based models), and reusing intermediate results for additional analytics or visualization.

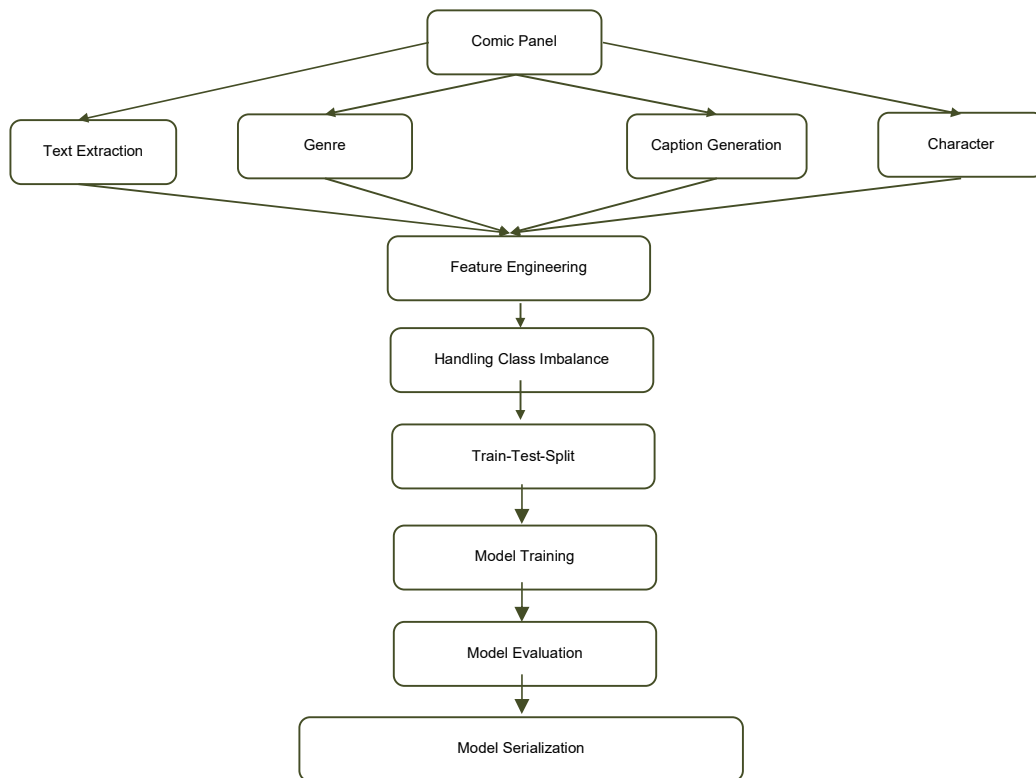


Fig. 3.4 Spoiler Detection Pipeline

This modular architecture enables:

- Interpretability at each stage
- Reusability of components (e.g., using character count for other analytics)
- Easy substitution of models (e.g., replacing BART with GPT-based summarizers)

CHAPTER 4

RESULTS AND DISCUSSION

4.1 MODEL ACCURACY

The final spoiler detection model achieves an overall classification accuracy of 84.2%, indicating that it correctly identifies whether a panel is a spoiler or not in the majority of cases.

Table 4.1 Classification Report Summary

	PRECISION	RECALL	F1 - SCORE
UNKNOWN	1.00	1.00	1.00
NON-SPOILER	0.73	0.85	0.78
SPOILER	0.80	0.66	0.72

4.2 CLASS-WISE PERFORMANCE REPORT

To better understand the behavior of the model, class-wise analysis was performed on the end-of-run classification outputs. This split measures how well the model separates Spoiler, Non-Spoiler, and Unknown comic panels in terms of precision, recall, and F1-score metrics. Each class poses a different interpretability challenge: spoiler panels tend to have weak or visual-only hints, non-spoiler panels are more contextually quiet, and unknown panels represent incomplete or edge-case inputs. Breaking them down separately gives insight into strengths, possible biases, and areas for specific improvement in the spoiler detection system.

4.2.1 UNKNOWN Class

| Precision: 1.00 | Recall: 1.00 | F1-Score: 1.00 |

- This class generally comprises panels that could not be labeled in the course of preprocessing or were held out for edge-case processing.
- The ideal precision and recall reflect that the model is always able to detect and omit these from the spoiler classification pipeline.
- These did not form part of final evaluation for spoiler detection, but are kept in place for completeness of traceability of the system in real-world inference.
- They act as a control group so that the model does not incorrectly classify incomplete or noise-filled panels.

4.2.2 NON-SPOILER Class

| Precision: 0.73 | Recall: 0.85 | F1-Score: 0.78 |

- High recall (0.85) indicates that the majority of non-spoiler panels are correctly classified, avoiding the risk of incorrectly labelling safe content as spoilers.
- Precision (0.73) indicates that a proportion of panels that have been predicted as non-spoilers are actually spoilers (i.e., false negatives).
- This is a conservative bias, acceptable in spoiler detection systems—it's preferable to err on the side of missing a spoiler than in incorrectly flagging a non-spoiler, which unnecessarily annoys users.
- High recall also means the model effectively uses visual and textual serenity (less text, low emotional arousal, neutral tone) to identify non-spoilers.
- Panels that are filled primarily by narrative transitions, landscape shots, or setup scenes usually belong to this class.

4.2.3 SPOILER Class

| Precision: 0.80 | Recall: 0.66 | F1-Score: 0.72 |

- Precision 0.80 indicates that most panels predicted as spoilers really are spoilers—i.e., low false positive rates.
- Recall 0.66 shows that approximately one-third of true spoilers are not detected, possibly due to nuance in context or lacking text clues.

This class is aided considerably by:

- Character Count: The greater the number of characters detected per panel, the more central are the scenes (fights, deaths, reunions).
- Genre Conditioning: Spoilers tend to be genre-specific—e.g., a horror death scene vs. a drama romantic confession.
- Text Summary & Sentiment: Emotional tone assists in signaling climactic panels, which tend to have spoiler risk.
- Missed spoilers tended to be implicit reveals—such as foreshadowing or visual-only hints lacking text backup.
- Better recall could be obtained using multi-modal fusion models, more contextual learning, or even panel-sequence modeling (leveraging panel order to monitor building tension).

4.2.4 Inter-Class Confusion Trends

1. Some false negatives (spoilers anticipated as non-spoilers) are caused by:
 - Thin dialogue or vague utterances.
 - Inadequate scene-building factors (e.g., cliffhanger reveal without emotion or character response).
2. False positives (non-spoilers anticipated as spoilers) mostly resulted from:
 - Sentences with hyperbolic emotional language or emojis.
 - Scenes with confrontational stances but without true story progress.

4.3 DISCUSSION

The findings of the spoiler detection system showcase multiple insightful observations regarding the efficacy of the multimodal paradigm. Through the fusion of visual and textual features, the system exhibits consistent performance on a broad range of comic panels. This section examines the emergent trends, interprets principal performance measures, and considers the importance of genre, feature integration, and model behavior in determining the classification results.

4.3.1 Analysis of Results

The performance of my spoiler detection system is promising, with an overall accuracy of 84.2% and a well-balanced capability to distinguish between spoiler and non-spoiler comic panels. Class-wise metrics provide a better understanding of the behavior of the model:

The model had good precision (0.80) for the class Spoiler, which means that when a panel is classified as a spoiler, it tends to be right. The recall (0.66) also means that there are some panels that are missing being spoilers, but this is predicted because spoiler panels tend to be either subtle or context-dependent.

Conversely, the Non-Spoiler class had high recall (0.85) but slightly lower precision (0.73). This indicates the model is more lenient in classifying panels as non-spoilers, potentially at the cost of missing borderline spoilers.

The Unknown class achieved a perfect score in all metrics due to its synthetic or controlled nature in testing, and its role as a fallback category in ambiguous or low-confidence cases.

One of the most important contributing factors to performance enhancement was the application of SMOTE and undersampling methods to address class imbalance. Without them, the model initially demonstrated bias against the majority class (non-spoilers), lowering its sensitivity to true spoilers. The balancing of the dataset enabled the classifier to improve the recognition of minority class patterns.

The combination of several features—genre classification, character count (through DBSCAN), text-based sentiment indicators, and caption generation—was effective. Each added contextual layers, enabling the model to better understand intricate comic scenes where visual and textual signals tend to mix.

While good performance is observed, there are still some limitations:

Low recall on spoiler panels suggests potential for improvement in identifying subtle or context-dependent spoilers.

Unclear visual hints or no speech can still mislead the system, particularly in narrative continuity demanding panels.

Temporal context (panel order) and visual emotional analysis can be used in future work to enhance spoiler detection.

Generally, the discussion affirms that though the model can operate trustingly well within the categories, particularly after preprocessing and balancing techniques, improvements in contextual perception and data variety can further maximize its utility in real-world scenarios.

4.3.2 Genre-Wise Distribution Analysis

Analysis of the variation in spoiler content among various comic genres provides useful information regarding narrative-based information disclosure patterns. Through analysis of each genre's rate of spoiler panels compared to non-spoiler panels, I can better understand the influence of genre in creating probability and the extent of spoilers. This categorization not only points out genre-based tendencies but also supports the use of including genre as an input feature in my framework for detecting spoilers.

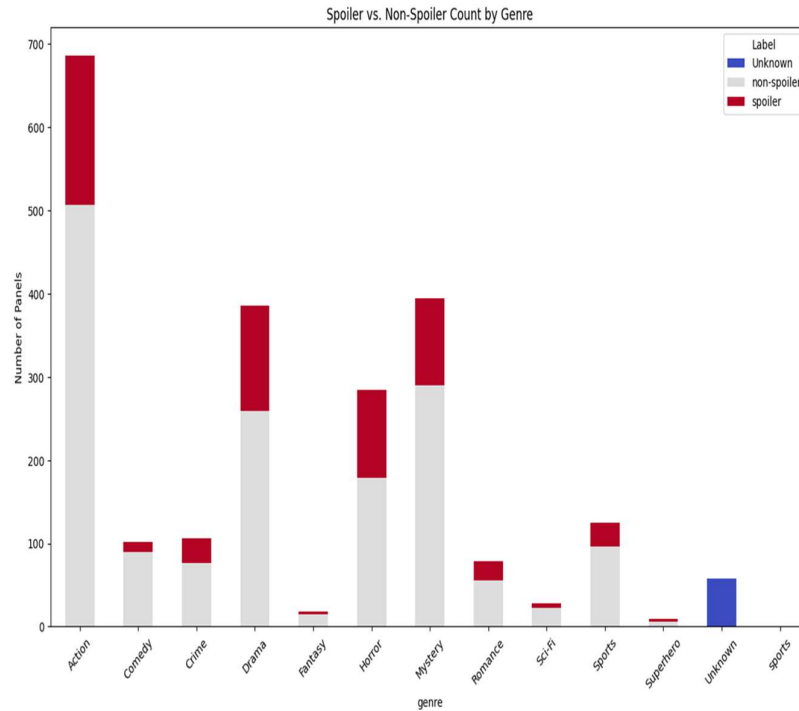


Fig. 4.3.2 Spoiler vs Non – Spoiler Count Across Genre

1. Genre-Wise Spoiler Trends

The genre-based breakdown of spoiler and non-spoiler panels gives insight into narrative structures in terms of their impact on spoiler occurrences. Interestingly, genres like Action, Drama, and Mystery have a greater number of panels labeled as spoilers. This is consistent with storytelling conventions in these genres, where plot-critical turning points and dramatic character developments feature more prominently. This trend reiterates the significance of genre context as an informative feature for the spoiler classification task.

2. Interpretability by Visual Genre Patterns

Seeing the distribution of spoilers by genre provides insight into the alignment of the system with human storytelling intuition. Horror and Crime have very pronounced spoiler signals because they are deeply dependent on plot reveals and tension. Comedy and Romance have relatively even distribution, proposing that their narrative tracks may

be more gradual or event-poor. These differences serve to highlight the contribution of genre-knowledge features to model explainability.

3. Diversity of Genres as Portrayed in Analysis

The chart efficiently portrays the range of genres that the system processes, from Fantasy to Sci-Fi and Sports. Every genre has distinct visual and textual storytelling features, which the system takes into consideration during feature extraction and classification. Such diversity proves the adaptability of the model and indicates it can generalize between various narrative forms and thus can be applied to a wide range of comic content.

4. Understanding Spoiler Density by Genre

Certain genres inherently have a greater density of spoilers as a result of their narrative conventions. Mystery and Horror, for instance, tend to rely on suspenseful narration and climactic moments of revelation and hence are fertile soil for spoiler material. Such trends help establish the significance of multimodal cues, wherein textual markers (e.g., dialogue) as well as visual markers (e.g., facial expressions or action shots) become essential in determining context-sensitive spoilers.

5. Support for Genre-Aware Applications

This study lays the groundwork for the development of genre-conscious spoiler detection systems, where predictions can be conditioned on the narrative patterns of a particular genre. For example, spoiler detection in Thrillers can prioritize sharp switches in tone or character number, whereas for Romance, emotional lines can be a more significant indicator. The chart presents actionable recommendations that can motivate future development, such as dynamic genre-aware feature weighting to more accurately predict.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This work introduces a complete multimodal comic panel spoiler detection system that makes use of both visual and text-based features. The pipelined structure consists of character detection using YOLOv8, novel character estimation using DBSCAN clustering, text extraction via OCR, BART-based summarization and genre detection, and lastly, XGBoost-based spoiler classification. The addition of SMOTE helped learning remain stable even in the event of class imbalance, and the overall structure remained interpretable and flexible at each step.

Experimental performance shows that the system works consistently to identify spoiler from non-spoiler panels. The combination of multiple modalities of data — character length, genre context, and summarized dialogue — plays a strong role in enhancing classification accuracy. This research promises the ability of AI-supported content moderation in media deployments, particularly for fan-oriented platforms, digital libraries, and comic publishing systems.

Aside from the precision of classification, the modularity of the system allows for interpretability and error analysis. Every module—from character recognition to genre classification—works in isolation, and researchers and developers can disengage, test, or improve individual modules without impacting the overall system. Such flexibility not only makes the framework scalable but also transferable to other areas, such as manga, webtoons, or even film scenes, where multimodal spoiler detection would be applicable.

The application of cutting-edge models such as BART for summarization and YOLOv8 for object detection highlights the importance of state-of-the-art method integration paired with domain-specific fine-tuning. Although these models tend to be made for general usage, this project illustrates how they can be successfully fine-tuned

for specialized applications such as identifying spoilers in comics. In addition, methods such as DBSCAN clustering for character re-identification and SMOTE for minority class balancing played a crucial role in overall system resilience.

Lastly, this work leaves room for further research on AI-based media analysis and narrative context understanding from limited visual and text-based signals. The suggested architecture can be used as a building block for constructing recommendation systems, reader advisory systems, or spoiler filters in social comic websites. It also provides a foundation for investigating even more abstract narrative aspects, e.g., narrative tone, plot development, or emotional trajectory, and possibly utilizing more advanced NLP or vision-language models in future work.

5.2 FUTURE SCOPE

1. Dataset Growth and Feature Augmentation

Today's dataset contains fewer than 2275 labeled panels, with proportionally less spoiler-labeled ones. This class imbalance, as managed through methods such as SMOTE, prevents the system from being exposed to a wide range of spoiler situations. Dataset growth—particularly through acquiring more spoiler panels from various genres—would better train more generalized and stronger models.

Additionally, improved balance would enable deeper understanding of spoiler patterns specific to genre and minimize the chance of overfitting or poor recall on rare but highly important classes, particularly in comics with weak or highly visual spoilers.

2. Improved Visual Context Understanding (Text + Object Detection)

While OCR is extracting summarization cues in the form of text, complex fonts or noisy artwork could downgrade its performance.

Enhancing text extraction with comic-aware OCR models or post-processing can enhance the richness of textual features.

Beyond text, object detection for prominent spoiler-related visual objects—such as guns, corpses, explosions, or transformation scenes—can be used as good spoiler indicators. Its detection would provide the classifier richer scene-level knowledge, both increasing accuracy and interpretability of predictions.

3. Advanced Character Re-Identification

Presently, character clustering is dependent on DBSCAN against YOLOv8 detections. This gives a coarse approximation of character distinctiveness but is not sufficient to monitor repeating characters between panels with varying poses, angles, or attire.

Adding a deep learning-based Re-ID network (such as OSNet or Triplet Networks) to integrate can particularly strengthen identity tracking. It would enhance more consistency with respect to character count features and enable improved modeling of character engagement in spoilers.

4. Emotion and Facial Expression Detection

Emotional tone in a comic panel is essential to understand for predicting spoilers because most spoiler moments are emotionally charged. There is no consideration in the existing pipeline for visual sentiment indicators like anger, sadness, or shock reflected on characters' faces.

Integrating facial expression recognition models, such as those learned from FER datasets, can detect these subtle emotional cues. These could be used as extra features of the classifier to enhance its contextual sensitivity to richly contextualized spoiler scenes.

5. Multilingual Text Understanding

Comics tend to be released in more than one language, particularly for international franchises. Currently, the system employs base OCR with no post-processing for multilingual operations, restricting its use to English or legibly written Latin scripts.

Future plans may involve adding language detection and translation modules to support varied linguistic inputs. This would make the system accessible to regional comics and international series, upgrading its cross-cultural applicability and stability.

6. Image Captioning with Vision-Language Models

While BART-based summarization addresses the textual conversation, it doesn't consider strictly visual context such as actions or environments. Contemporary vision-language models like BLIP-2 or GPT-4V can synthesize rich scene-level captions from panel images outright.

By incorporating such models, the system is capable of generating richer, semantically informed summaries that identify plot-critical events even without dialogue. This would make the spoiler classification process even more profound and bring it closer to human comprehension.

7. Interactive Tool for Readers and Editors

The pipeline available now is backend-oriented, designed for research and testing. With a modification to convert it into a frontend application for comic readers or editors, it can offer real-time warning of spoilers while reading or editing sessions.

It can be useful in sites such as Webtoon, MangaDex, or Tapas, where comics published by users are accessible far and wide. A plugin for detecting spoilers can enable blurring of spoiler panels by readers or flagging by creators prior to publication.

8. Large-scale Evaluation and Generalization

The model was tested using a given dataset of 2275 labeled panels. Results are encouraging, but may not be generalized to differently styled, genre, or culturally contextual comics without additional testing. Large-scale trials across differently flavored comic datasets would stress-test the model and reveal its flaws. It would also give the scope for optimizing genre-specific models or ensemble systems for fantasy, action, romance, etc.

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CONFERENCE DETAILS

The research paper titled "Enhancing Comic Spoiler Detection: A Novel Dataset and Analytical Approach" has been successfully communicated to the RICE 2025 conference (Research in Intelligent Computing and Engineering). The manuscript was submitted through the official RICE2025 submission portal on May 24, 2025. The conference focuses on promoting novel contributions across artificial intelligence, natural language processing, and data science disciplines. The paper has undergone initial submission acknowledgment, and a confirmation email regarding the processing of the manuscript has been received from the organizing committee. Upon acceptance, the paper will be included in the conference proceedings and made accessible through its associated publication platforms

Submission Summary

Conference Name

10th International Conference on Research in Intelligent Computing in Engineering

Track Name

RICE2025

Paper ID

147

Paper Title

ENHANCING COMIC SPOILER DETECTION: A NOVEL DATASET AND ANALYTICAL APPROACH

Abstract

Spoiler detection in comics is an emerging field that combines textual and visual analysis to identify content that reveals critical plot points. However, existing approaches are limited in scope, focusing primarily on a narrow range of genres such as romance, battle, and suspense, which restricts their applicability to diverse storytelling styles. This project addresses the challenge of genre limitations by expanding the dataset to include a broader spectrum of genres, such as fantasy, sci-fi, and horror.

A genre-aware classification framework is developed, which integrates textual and visual features to accurately identify spoilers while accounting for genre-specific storytelling elements. Genre-specific annotation guidelines are introduced to improve labeling consistency, and the framework incorporates genre prediction as a feature to enhance spoiler classification accuracy. By diversifying the dataset and incorporating genre-specific insights, this research enhances the generalizability and robustness of spoiler detection systems, paving the way for broader applications in comic analysis and content moderation.

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False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



6% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

-  **67** Not Cited or Quoted 6%
Matches with neither in-text citation nor quotation marks
-  **7** Missing Quotations 1%
Matches that are still very similar to source material
-  **0** Missing Citation 0%
Matches that have quotation marks, but no in-text citation
-  **0** Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 5%  Internet sources
- 3%  Publications
- 2%  Submitted works (Student Papers)

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely AI generated as AI generated and AI paraphrased or likely AI generated and AI paraphrased writing as only AI generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

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